# 4 Field Effect Transistors

## FIELD EFFECT TRANSISTORS (FET)

### Introduction to FET

- The FETs are categorized into 2 parts:
  **1)** JFET or the junction field effect transistors.
  **2)** MOSFET or metal oxide semiconductor field effect transistors.
- The operation of FET is controlled by the electric field intensity produced in the device.
- FET is a voltage controlled and unipolar device.
- The input resistance is very high (in Mega ohms) in FET.
- No minority carriers are present in this device; it is a majority carrier device.
- This device has excellent thermal stability; hence no leakage current or the effect of temperature is seen in them.
- Negligible power dissipation is observed in this device.
- Cut-in voltage in FET is zero.

**Junction field effect transistor:**

- JFET includes three terminals:

**Source (S):**

- The source terminal is the source of the majority carriers.

**Drain (D):**

- The drain terminal in JFET drains off the majority carriers in the device.

**Gate (G):**

- The gate terminal controls the majority of carries that are coming from the source and reaching the drain terminal.

**Construction and working of JFET:**

- The 2 types of JFET, depending upon the channel created, are:

**Grey Matter Alert!**

As the magnitude of gate voltage increases, the channel width is reduced. Now, fewer majority carriers reach the drain terminal hence reducing the value of the drain current ($I_D$).

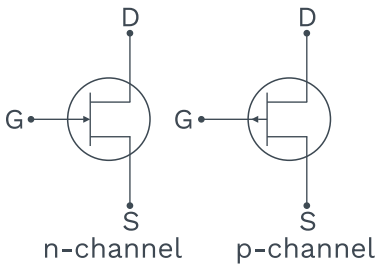



**Fig. 4.1 Symbols of JFET**

### 1) N-Channel JFET

- Construction:

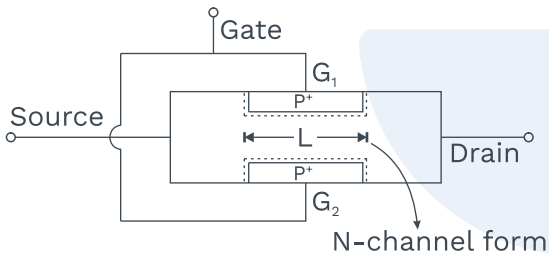



**Fig. 4.2 Construction of N-Channel JFET**

- The gate voltage ($V_{GS}$) applied in n-channel JFET is negative.
- The gate to the source is always operated under reverse biased conditions.

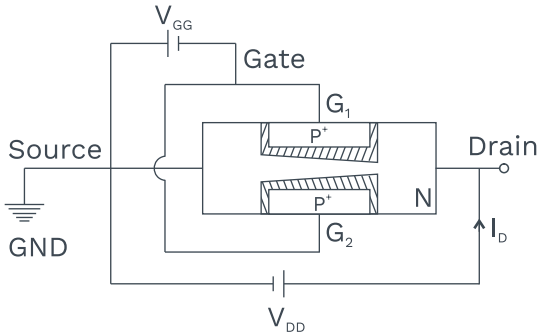



**Fig. 4.3 Working of n-channel JFET**

- The current $I_D$ flows through the N-type channel passing between the two p type regions. The electrons drift from source to drain i.e. left to right. And the flow is opposite to the direction of the current.

**Grey Matter Alert!**

The emitter-base junction breakdown occurs due to the Zener effect and the collector-base junction breakdown occurs due to the avalanche effect.

**Previous Years' Question**

In a JFET if

| | |
|---|---|
| **a)** the pinch-off voltage decreases. | **1)** the channel doping is reduced. |
| **b)** the transconductance increases. | **2)** the channel length is increased. |
| **c)** the transit time of the carriers | **3)** the conductivity of the channel |
| **d)** the channel length is reduced. | in the channel is reduced and increased. |
| | **4)** the gate area is reduced. |

- The reverse bias between the two $P^+$ depletion regions and the channel makes the depletion region invade the N-region.
- Depletion region penetrates more near the drain terminal and lesser near the source terminal into the channel.
- As the value of $V_{DS}$ is increased, the channel width reduces. Channel potential varies along the length and the channel are wedge-shaped.

**2) P-Channel JFET**

- Construction:

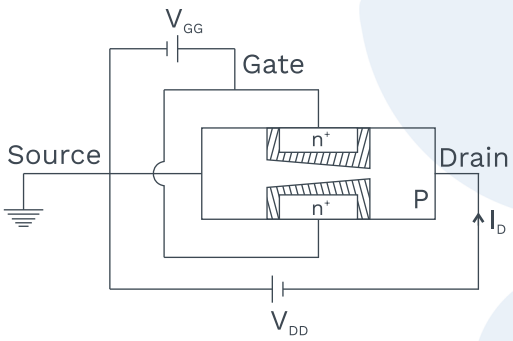



**Fig. 4.4 Working of P-Channel JFET**

- For a p-channel JFET, the gate voltage($V_{GS}$) is positive.
- Holes flow from the source to the drain; it is in the same direction as the current $I_D$.
- Gate regions are $n^+$ in the P-channel JFET.
- The majority of carriers in a JFET are dependent upon.
- Cross-sectional area of the channel.
- Doping concentration of the channel.

**Grey Matter Alert!**

FET can also work as a voltage variable resistor by the variation in gate to source voltage ($V_{GS}$).

**Transfer characteristics of FET:**

- Based upon the polarity of the gate to the source voltage, the transfer characteristics are plotted either in the 1st or 2nd quadrant.
- The transfer characteristics are plotted for saturation region only; in which the FET is

**Grey Matter Alert!**

The ohmic region is also called the linear region or active region or non-saturation region.

generally operated.

**Step 1:**

- Keep $V_{DS}$ (drain to source voltage) constant.
- Vary $V_{GS}$ (gate to source voltage) and the drain current are noted.
- At the cut-off value of gate to source voltage, the two depletion layers become extremely close to each other but do not touch. Hence, no majority carries to reach the drain terminal. This is the point where the drain current reduces to zero.
- $I_D$ versus $V_{GS}$ keeping $V_{DS}$ constant.

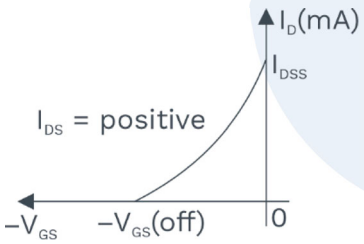



**Fig. 4.5 Transfer Characteristics of n-Channel FET**

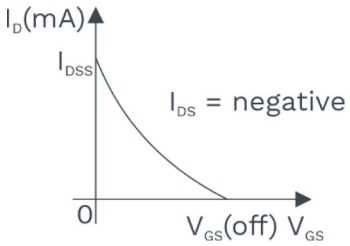



**Fig. 4.6 Transfer Characteristics of p-Channel FET**

- $I_{DSS}$ is the drain to source saturation current obtained when $V_{GS}$ = 0; $I_D$ = $I_{DSS}$.
- As evident from the graph, drain current $I_D$ reduces as the gate to source voltage becomes more negative.
- The minimum gate to source voltage required to reduce the value of drain current to zero is called $V_{GS}$ (off) or the threshold voltage.

**Step 2:**

- Keep $V_{GS}$ (gate to source voltage) constant.
- Vary $V_{DS}$ (drain to source voltage), and the

**Grey Matter Alert!**

FET can work as a constant current source in the pinch-off region.

**Rack Your Brain**

Can FET work as an amplifier? If yes, in which region?

**Rack Your Brain**

Who proposed the drain current equation?
Hint: The drain current and the diode current equation is famous by this name.

drain current is noted.

$I_D$ versus $V_{DS}$ keeping $V_{GS}$ = constant.

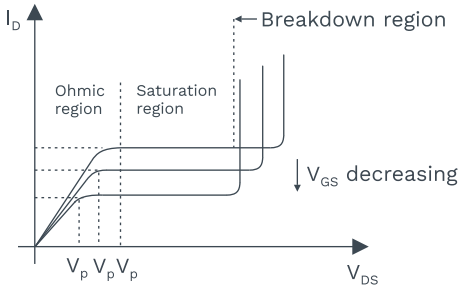



**Fig. 4.7 Regions in the FET**

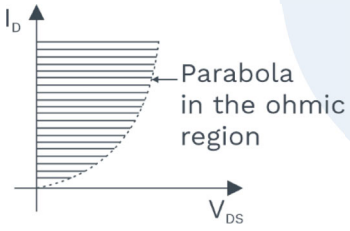



**Fig. 4.8 Ohmic Region in the FET**

**Ohmic, saturation and breakdown regions:**

- In the **'ohmic region'**, the channel behaves as a resistor.
- The FET works as a linear device and follows a parabolic characteristic in this region.
- The ohmic and saturation regions are separated by a parabola.
- The **saturation region** is also called a pinch-off region. After the pinch-off voltage ($V_p$), the drain current is constant.
- FET is usually operated in the saturation region only and is also called square-law device because of its characteristics in this region.

**Pinch off voltage ($V_p$):**

- It is defined as the minimum value of drain to source, voltage required to make the drain current ($I_D$) enter the saturation.
- The pinch-off voltage decreases as the gate

**Previous Years' Question**

An n-channel JFET has $I_{DSS}$ = 1 mA and $V_P$ = –5V. It's maximum transconductance is_____.

to source voltage is increased.

**Drain current in saturation:**

$$I_D = I_{DSS}\left[1-\left|\frac{V_{GS}}{V_p}\right|\right]^2$$

- $I_D$ is a majority carrier current.
- The drain current '$I_D$' is a drift current as it is passing through the channel under the influence of an electric field.
- It is dependent upon temperature.
- For 1-degree increase in temperature, the drain current decreases by 0.7%.
- $I_D \propto T^{-3/2}$

**Transconductance ($g_m$):**

- The most important parameter of FET is transconductance / mutual conductance ($g_m$).

By definition:

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}} \text{ or } \left.\frac{\Delta I_D}{\Delta V_{GS}}\right|_{V_{DS}} \text{ ʊ}$$

By the drain current equation of FET in saturation:

$$\Rightarrow I_D = I_{DSS}\left[1-\left|\frac{V_{GS}}{V_P}\right|\right]^2 \quad .....1)$$

Differentiating both the sides:

$$\Rightarrow \frac{\partial I_D}{\partial V_{GS}} = 2I_{DSS}\left[1-\left|\frac{V_{GS}}{V_P}\right|\right]\left[\frac{-1}{V_P}\right]$$

$$\Rightarrow g_m = \frac{-2I_{DSS}}{V_P}\left[1-\left|\frac{V_{GS}}{V_P}\right|\right] \text{ ʊ} \quad .....2)$$

From equation 1):

$$\Rightarrow 1-\left|\frac{V_{GS}}{V_P}\right| = \sqrt{\frac{I_D}{I_{DSS}}}$$

Equation 2) becomes:

$$\Rightarrow g_m = \frac{-2}{V_P}\sqrt{I_D . I_{DSS}}$$

Let $I_D = I_{DS}$

**Grey Matter Alert!**

Drain characteristics of FET are called constant current characteristics also. They are analogous to the collector characteristics of common base BJT.

$$g_m = \frac{-2}{|V_p|}\sqrt{I_{DS}.I_{DSS}}$$

When $V_{GS} = 0 \rightarrow I_{DS} = I_{DSS}$

- $g_m = \frac{-2I_{DSS}}{V_p}\mho$
- This is the maximum value of transconductance of FET.

**Other FET parameters:**

**1) Drain-source resistance**

$$r_{ds} = \frac{V_{DS}}{I_D}\Omega$$

Now, consider the cross-sectional view of an n-channel JFET:

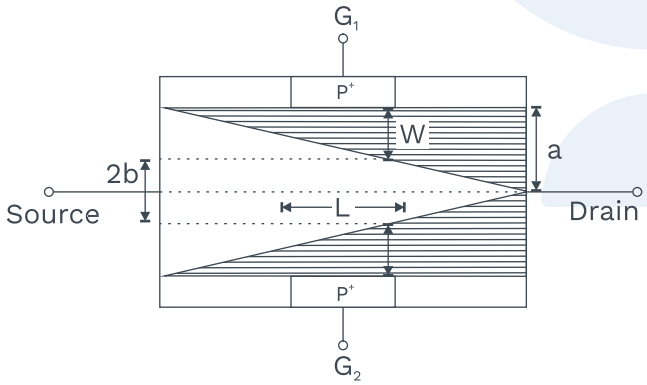



**Fig. 4.9 Cross-Sectional View of JFET**

Where

- W is the width of the depletion layer on each side.
- 2b is the channel width near drain.
- L is the channel length.
- a is the half channel width.

**2) Pinch off voltage**

$$|V_p| = \frac{qN_Da^2}{2\epsilon} \text{ Volts}$$

**Grey Matter Alert!**

Space charged or neutral charge means for every hole, there is a negative charge also i.e. immobile.

**Previous Years' Question**

Consider the following statements S1 and S2.

**S1:** The threshold voltage ($V_T$) of a MOS capacitor decreases with increase in gate oxide thickness.

**S2:** The threshold voltage ($V_T$) of a MOS capacitor decreases with increase in substrate doping concentration.

Which one of the following is correct?

**a)** S1 is FALSE, and B2 is TRUE.

**b)** Both S1 and S2 are TRUE.

**c)** Both S1 and S2 are FALSE.

**d)** S1 is TRUE, and S2 is FALSE.

Where $\in = 11.7\in_0$

- $V_p$ is positive for n-channel JFET and negative for p-channel JFET.

**3) Drain current equation**

$$I_D = 2b\,qN_D\mu_n\left[\frac{V_{DS}\cdot W}{L}\right]$$

Let $\frac{V_{DS}}{L}$ = electric field (E)

$I_D = 2b\ qN_D\mu_n\ WE$

Let 2bW = Cross sectional area of the channel

$I_D = AqN_D\mu_n E$

**4) Relation between the terminal voltages**

$V_{DS} = V_{GS} + V_{DG}$

& $|I_S| = |I_D|$

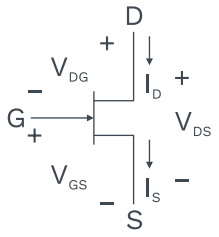



**Fig. 4.10 Biasing of JFET**

**Example 1:**

Consider the circuit in the figure below:

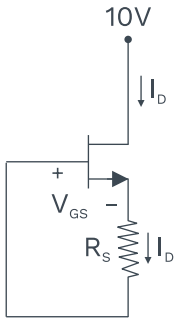
Assume:

$I_{DSS} = 10mA$

$V_p = -5V$

$I_D = 6.4mA$

Find $R_s$.




**Sol:** Using the equation:

$$I_D = I_{DSS}\left(1-\frac{V_{GS}}{V_p}\right)^2$$

$$\Rightarrow\quad V_{GS} = V_p\left(1-\sqrt{\frac{I_D}{I_{DSS}}}\right)$$

**Previous Years' Question**

A MOS capacitor made using p-type substrate is in the accumulation mode. The dominate charge in the channel is due to the presence of

**a)** holes

**b)** electrons

**c)** positively charged ions

**d)** negatively charged ions

**Rack Your Brain**

Can the metallic plate used in MOS-Cap be made up of poly-crystalline silicon?

$$= -5\left[1 - \sqrt{\frac{6.4 \times 10^{-3}}{10 \times 10^{-3}}}\right] = -1V$$

$$V_{GS} = -I_D R_S$$

$$R_s = \frac{1}{6.4 \times 10^{-3}} = 156.25\,\Omega$$

**MOSFET (Metal Oxide Semiconductor Field Effect Transistor)**

**Construction and Working:**

- MOS-Cap (MOS-Capacitor) or Metal Oxide Semiconductor capacitor is the heart of MOSFET.
- The basic structure of a MOS-Cap is:

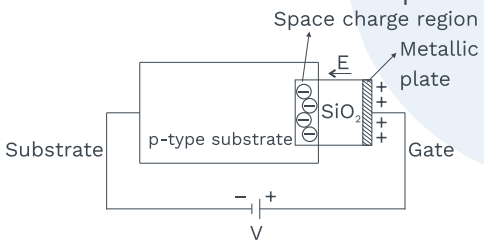



**Fig. 4.11 MOS Capacitor with p-Type Substrate**

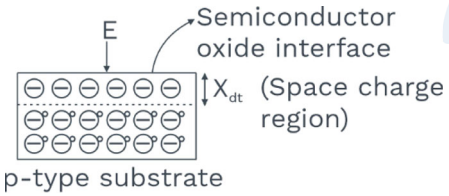



**Fig. 4.12 States of Substrate in the MOS Capacitor**

- Substrate can be n-type or p-type.
- Metallic plate can be made up of silver or aluminium.
- $t_{ox}$ is the thickness of the oxide.

**Working:**

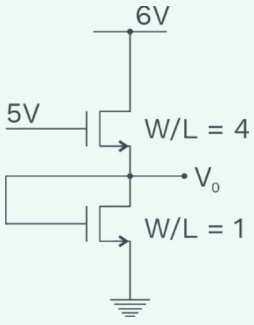
- The upper plate of the MOS capacitor is biased negatively with respect to the plate at the bottom of the capacitor.
- Electric field is directed in the upwards direction.
- This happens because the negative charge gets deposited at the upper plate & the positive charge at the bottom plate.

**Grey Matter Alert!**

$\in_r(Si) = 11.7$
$\in_r(SiO_2) = 3.9$

**Previous Years' Question**

In the circuit shown below, for the MOS transistors, $\mu_n C_{ox}$ = 100 mA/V$^2$ and the threshold voltage $V_T$ = 1V. The voltage $V_x$ at the source of the upper transistor is:




**a)** 1 V **b)** 2 V
**c)** 3 V **d)** 3.67 V

**Depletion mode:**

- Gate is positively biased with respect to the substrate.
- The electric field intensity penetrates the interface.
- Majority of carriers in the substrate (holes) move away from the interface.
- A region of immobile negative ions called the space charge region is created.

**Inversion mode:**

- When the gate voltage is very large, the electric field attracts even the minority carriers in the p-type substrate.
- Due to this, region of negative charges called the inversion layer is formed when the electrons deposit in the space charge region.

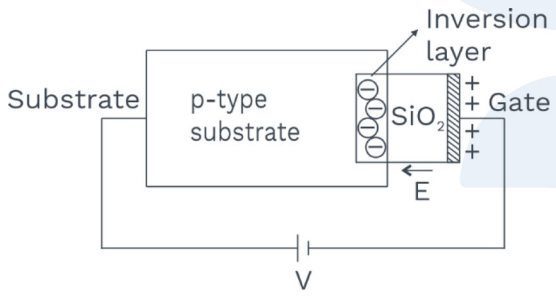



**Fig. 4.13 Inversion Mode in MOS-Cap**

**Rack Your Brain**

What is the name of the transistor formed when the gate terminal is isolated by $SiO_2$?

## ACCUMULATION MODE

- If the gate is negatively biased with respect to the substrate, as in the figure the majority carriers of the p-type substrate accumulate near the interface between the $SiO_2$ layer and the substrate. Hence, holes are forced to move towards the semiconductor-oxide interface.

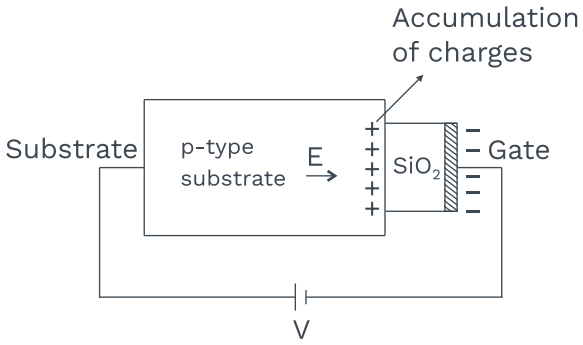



**Fig. 4.14 Accumulation Mode in MOS-Cap**

**Flat band voltage:**

- The voltage developed at the semiconductor oxide interface in a MOS capacitor is known as flat band voltage ($V_{FB}$).
- Consider: $\phi_{ms}$ as the metal-semiconductor work function in volts:
  $\phi_{ms} = \phi_m - \phi_s$
- $C_{ox}$ is the oxide capacitance given as:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

Where
$t_{ox}$ is the thickness of the oxide.
$\epsilon_{ox} = \epsilon_r \epsilon_o$, which is the permittivity of oxide.
$\epsilon_r$ is the relative permittivity of the material.
$\epsilon_o$ is the permittivity of vacuum.

- $Q_{ox}$ is the interface charge given as:

$Q_{ox} = Q'_{ss} \times q$
where $Q'_{ss}$ is the transferred charge present at the semiconductor oxide interface.
Then

$$V_{FB} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}}$$

- Threshold voltage of MOSFET is:
  $V_T = V_{FB} + 2\phi_F$

Where $\phi_F$ is the fermi voltage.

**Comparison between MOSFET and JFET:**

**Table: Difference between JFET and MOSFET**

| MOSFET | JFET |
|---|---|
| **1)** MOSFET is operated in depletion and enhancement modes.<br>**2)** The input resistance is very high. (>10,000 mega ohms)<br>**3)** The gate is insulated from the channel by the $SiO_2$ layer.<br>**4)** Channel exists in depletion type only. | **1)** JFET is operated in depletion modes.<br>**2)** Input resistance is very high. (>10 mega ohms)<br>**3)** The gate is not insulated.<br>**4)** Channel exists in JFET permanently. |

**Table 4.1**

## TYPES OF MOSFETS

- The 2 types of MOSFET depend upon the channel created are:
  **1)** Depletion Type MOSFET (D-MOSFET)
  **2)** Enhancement Type MOSFET (E-MOSFET)

**E-MOSFET**

**Table: Difference between D-MOSFET and**

| Depletion Type Mosfet (D-MOSFET) | Enhancement Type Mosfet (E-MOSFET) |
|---|---|
| • Suitable to operate both in depletion mode and enhancement mode.<br>• D-MOSFET is sometimes called depletion/enhancement MOSFET.<br>• Pre-existing channel is available.<br>• Types of D-MOSFET<br>**i)** N-channel depletion MOSFET<br>**ii)** P-channel depletion MOSFET | • Suitable to operate only in enhancement mode.<br>• E-MOSFET is sometimes called E-only MOSFET.<br>• There is no pre-existing channel.<br>• Types of E-MOSFET<br>**i)** N-channel enhancement MOSFET<br>**ii)** P-channel enhancement MOSFET |

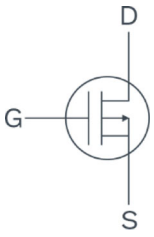
**Table 4.2**

**Symbols of different types of MOSFETS:**

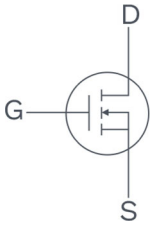
- An arrow pointing in represents an N-channel device, while an arrow pointing out represents a P-channel device.



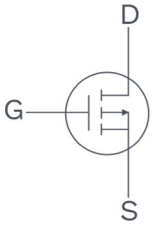

**a) Depletion Type NMOS**




**b) Depletion Type PMOS**




**c) Enhancement Type NMOS**




**d) Enhancement Type PMOS**

**Fig. 4.15 Symbols of Different Types of MOSFETS**

### Previous Years' Question

At room temperature, a possible value for the mobility of electrons in the inversion layer of a silicon n-channel MOSFET is:

**a)** 450 $cm^2/V\text{-}s$

**b)** 1350 $cm^2/V\text{-}s$

**c)** 1800 $cm^2/V\text{-}s$

**d)** 3600 $cm^2/V\text{-}s$

### Definitions

Channel width modulation:
The process in which the variation in the gate to source voltage alters the width of the channel.

## CONSTRUCTION, WORKING AND CHARACTERISTICS OF MOSFETS

**1) D-MOSFET:**

- As we can see, the D-MOSFET have physical channel between the source and drain terminals.

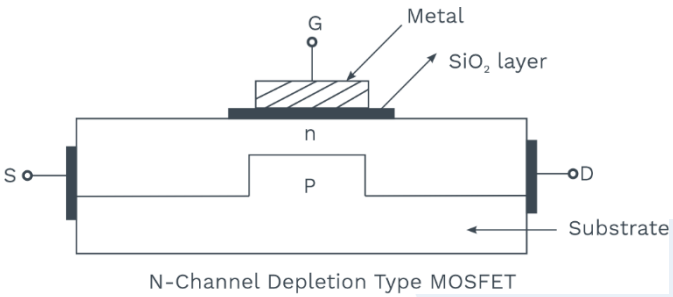



**Fig. 4.16 Construction of N-CHANNEL D-MOSFET**

- Consider the D-MOSFET operating conditions with gate and source terminals shorted together($V_{GS}$=0V).
- With voltage $V_{DS}$, an appreciable drain current $I_{DSS}$ flows. $I_D = I_{DSS}$
- $V_{GS}$ < 0: Positive charges are induced in the channel through the $SiO_2$ of the gate capacitor.
- As $V_{GS}$ is made more negative, the induced positive charges make the channel less conductive, and the drain current decreases.
- Re-distribution of charge in the channel causes a depletion of majority carriers - biasing voltage $V_{GS}$ depletes the channel of free carriers. It reduces the width of the channel, hence increasing its resistance. As a result, $I_D < I_{DSS}$
- The actual value of $I_D$ depends on the value of $I_{DSS}$, $V_{GS(off)}$ and $V_{GS}$.
- $V_{GS}$>0, the channel is widened. This reduces the resistance of the channel. And $I_D > I_{DSS}$
- The holes in the p-type substrate are repelled by the positive gate voltage, minority carriers in the p-type material are attracted towards the channel by the gate voltage.

**Previous Years' Question**

For an n-channel enhancement type MOSFET, if the source is connected at a higher potential than that of the bulk (i.e. $V_{SB}$ > 0), the threshold voltage $V_T$ of the MOSFET will

**a)** remain unchanged
**b)** decrease
**c)** change polarity
**d)** increase

- Electrons accumulate in the area to the right of the physical channel effectively becomes an N-type material hence allowing more drain current.

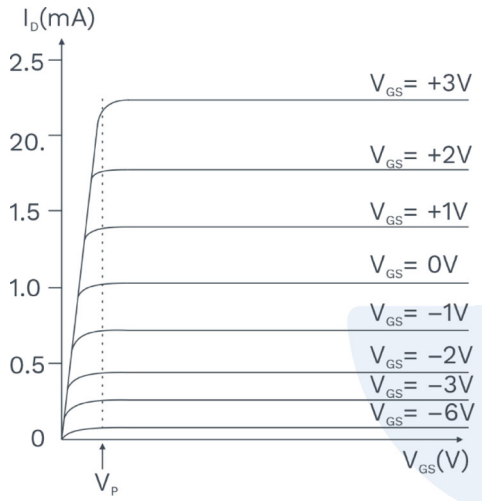



**Fig. 4.17 Drain Characteristics of MOSFET**

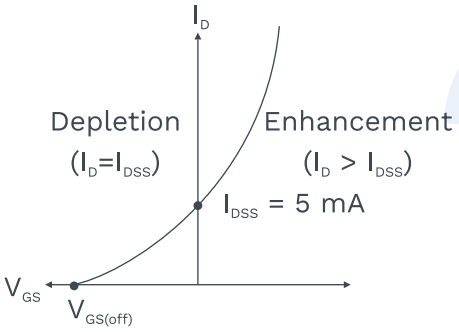



**Fig. 4.18 Transfer Characteristics of D-MOSFET**

- The above curves are similar to JFET, so the D MOSFET have the same transconductance equation.

**2) E-MOSFET:**

- The E-MOSFET, on the other hand, has no such channel physically.
- It depends on the gate voltage to form a channel between the source and the drain terminals.

**Previous Years' Question**

Consider the following two statements about the internal conditions in an n-channel MOSFET operating in the active region.

**S1:** The inversion charge decreases from source to drain.

**S2:** The channel potential increases from source to drain.

Which of the following is correct?

**a)** Only S2 is true.

**b)** Both S1 and S2 are false.

**c)** both S1 and S2 are true, but S2 is not a reason for S1.

**d)** Both S1 and S2 are true, and S2 is a reason for S1.

- These are initially switched off devices.
- Very Good Switches, so used in digital circuits.
- Mostly source connected to the body to reduce the number of independent sources.

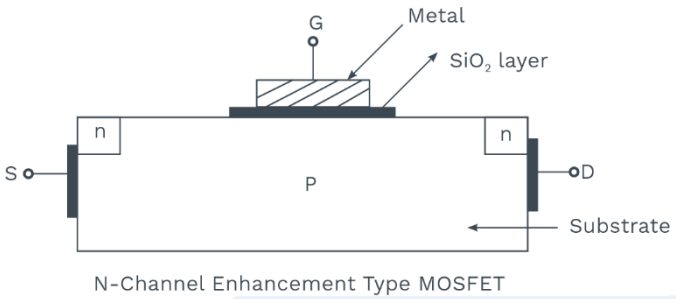



**Fig. 4.19 Construction of N-CHANNEL E-MOSFET**

- When the value of $V_{GS}$ =0 V, there is no channel connecting the source and drain materials, so there is no significant amount of drain current.
- The presence of the p-region does not permit the electrons coming from the source terminal to pass through it.
- $V_{GS} > 0$ , a negative charge is induced in the p-type substrate adjacent to the silicon dioxide layer.
- The current $I_D$ increases slowly and then much more rapidly.
- N-type bridge between source and drain is formed because holes are repelled, and electrons are attracted towards this voltage.
- The minimum $V_{GS}$ voltage which produces this inversion layer is called threshold voltage ($V_{GS(TH)}$). This is the point at which the device turns on.
- The standard transconductance formula will not work for the E MOSFET.
- To determine the value of $I_D$:
- $I_D = K[V_{GS} - V_{GS(TH)}]^2$

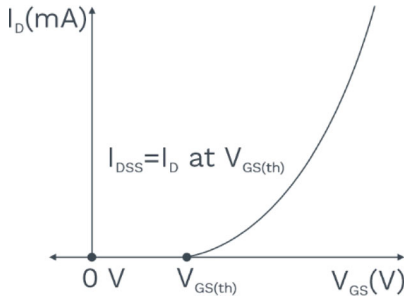



**Fig. 4.20 Transfer characteristics of E-MOSFET**

**Example 2:**

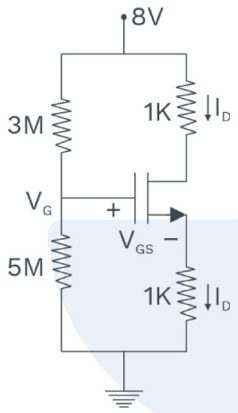
Consider the circuit in the figure below:

Assume:

$K_n = 1\ mA/V^2$

$V_{th} = 1\ V$

Find $(V_{DS}, I_D)$.




**Sol:**

By voltage divider rule:

$$V_G = \frac{8 \times 5M}{3M + 5M} = 5V$$

And $5 = V_{GS} + I_D \times 1\ K$ ..... 1)

Assume MOSFET is a saturation region.

$$I_D = \frac{1}{2} K_n (V_{GS} - V_{th})^2$$

$I_D$ is calculated from equation 1):

$$\frac{5 - V_{GS}}{1 \times 10^3} = \frac{1}{2} 10^{-3} [V_{GS} - 1]^2$$

$10 - 2V_{GS} = V^2_{GS} - 2V_{GS} + 1$

$\Rightarrow V_{GS}^2 = 9$

$\Rightarrow V_{GS} = \pm 3$Volts

Hence: $V_{GS1} = 3V$, $V_{GS2} = -3\ V$ is not possible as

It is less than the threshold voltage $(V_{Th})$.

From equation 1) and the value of

$V_{GS1} = V_{GS} = 3V$: $I_D = 2\ mA$

And

$V_{DS} = V_{DD} - I_D(R_D + R_S)$

$= 8 - 2 \times 2 = 4V$

$4 > V_{GS} - V_{Th}$

$\Rightarrow$ $4 > 3 - 1$

$\Rightarrow$ Hence, transistor is in saturation.

**Example 3:**

Consider the circuit in the figure below:

Assume:

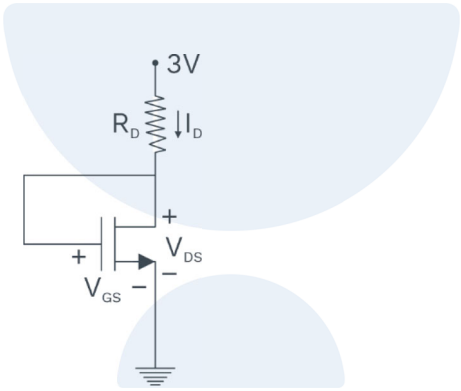
$V_{Th} = 0.6V$

$K'_n = 0.2\ mA/V^2$

$W/L = 5$

$I_D = 80\,\mu A$

Find $R_D$.




**Sol:**

As $V_{DS} = V_{GS} \Rightarrow$ transistor is in saturation

$$I_D = \frac{1}{2}K'_n\left(\frac{W}{L}\right)(V_{GS} - V_{Th})^2$$

$$80 \times 10^{-6} = \frac{1}{2} \times 0.2 \times 10^{-3} \times 5[V_{GS} - 0.6]^2$$

$\Rightarrow$ $V_{GS} = 1V = V_{DS}$

$$R_D = \frac{3 - V_{DS}}{I_D} = \frac{3 - 1}{80 \times 10^{-6}} = 25\,k\Omega$$

**Example 4:**

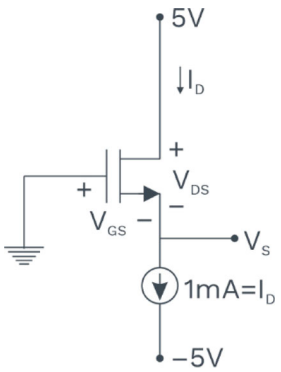
Consider the circuit in the figure below:

Assume:

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right) = 0.5mA\ /\ V^2$$

$V_{Th} = 1.2V$

Find $V_{DS}$.




**Sol:**

$V_D = 5V > V_G [= 0V]$

$\Rightarrow$ So, the transistor is in saturation region.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)\cdot[V_{GS} - V_{Th}]^2$$

$$1\times 10^{-3} = \frac{1}{2}\times 10^{-3}[V_{GS} - 1.2]^2$$

$$\sqrt{2} + 1.2 = V_{GS}$$

$$2.614V = V_{GS}$$

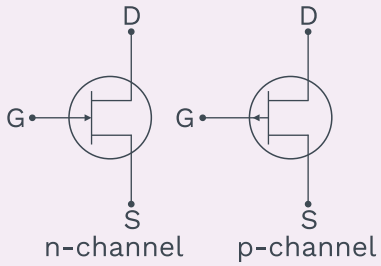
Now,

$V_S = -V_{GS} = -2.614V$

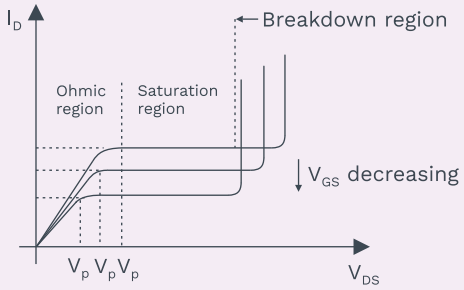
$V_{DS} = V_D - V_S = 5-(-2.614)= 7.614V$

## Chapter Summary

- The FETs are categorized into 2-parts:
  **1)** JFET or the junction field effect transistors.
  **2)** MOSFET or metal oxide semiconductor field effect transistors.
- JFET includes three terminals:
  **1)** Source (S)
  **2)** Drain (D)
  **3)** Gate (G)




- The gate to the source is always operated under reverse biased conditions.




- **Pinch off voltage:** The minimum value of drain to source voltage required to make the drain current ($I_D$) enters the saturation region.

$$I_D = I_{DSS}\left[1 - \left|\frac{V_{GS}}{V_p}\right|\right]^2$$

- For 1°C increase in temperature, the drain current decreases by 0.7%.
  $I_D \propto T^{-3/2}$

$$g_m = \frac{\partial I_D}{\partial V_{GS}}\bigg|_{V_{DS}} \text{ or } \frac{\Delta I_D}{\Delta V_{GS}}\bigg|_{V_{DS}} \text{ ʊ}$$

- The maximum value of transconductance of FET.

$$g_m = \frac{-2I_{DSS}}{V_p}\mho$$

$$r_{ds} = \frac{V_{DS}}{I_D}\Omega$$

$$|V_p| = \frac{qN_Da^2}{2\in} \text{ Volts}$$

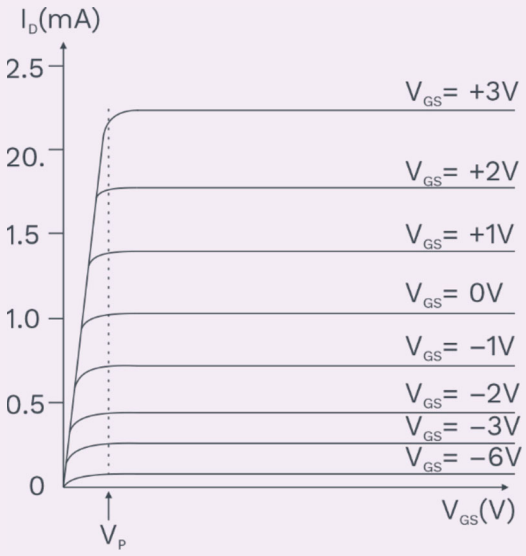
$$I_D = AqN_D\mu_nE$$

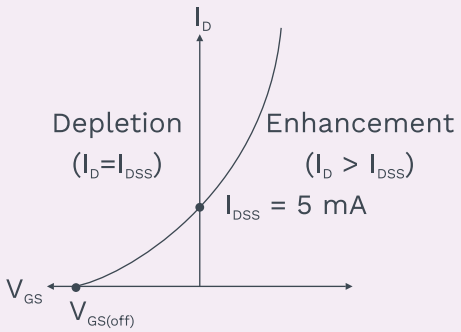
- Metal Oxide Semiconductor capacitor is the heart of MOSFET.
- A region of immobile negative ions called the space charge region is created.
- The voltage developed at the semiconductor oxide interface in a MOS capacitor is known as flat band voltage ($V_{FB}$).

$$V_{FB} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}}$$

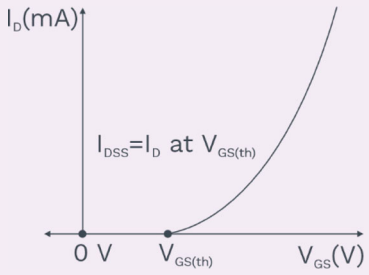
$$V_T = V_{FB} + 2\phi_F$$

- The 2-types of MOSFET, depending upon the channel created, are:
  **1)** Depletion type MOSFET (D-MOSFET)
  **2)** Enhancement type MOSFET (E-MOSFET)
- D-MOSFET have a physical channel between the source and drain terminals.






- The E-MOSFET, on the other hand, has no such channel physically.
- The standard transconductance formula will not work for the E-MOSFET.
  $I_D = K[V_{GS} - V_{GS(TH)}]^2$



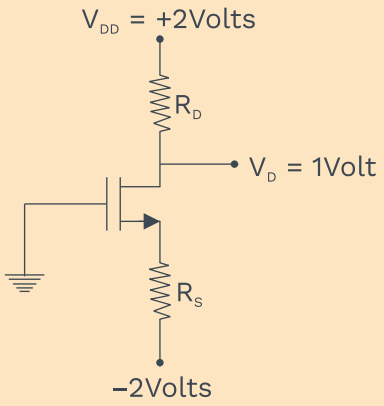

- The minimum $V_{GS}$ voltage which produces this inversion layer is called threshold voltage ($V_{GS(TH)}$).

## PRACTICE QUESTIONS

**Q1** **Which region will have the highest resistivity in the depletion-type MOSFET?**

**a) Source** **b) Substrate** **c) Drain** **d) Gate**

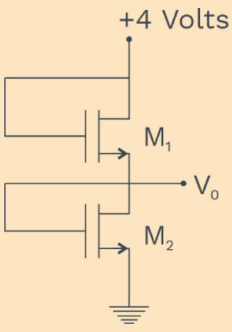
**Q2** **Find out the region in which the NMOS given in the circuit below is operating:**




**a) Saturation** **b) Triode**

**c) Cut-off** **d) Can't be calculated**

**Q3** **Consider the circuit shown in the figure below.**




**For both the transistors, consider:**

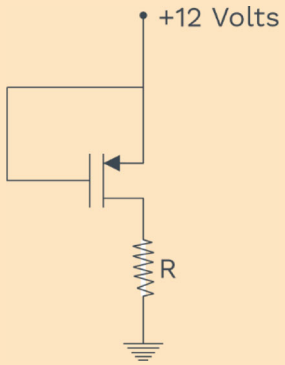
**$V_{Th}$ = 0.7V, $k_n'$ = 35μA/V²**

What is the value of $V_0$ (output voltage) if the ratios are given as:

$$\left(\frac{W}{L}\right)_1 = 45, \left(\frac{W}{L}\right)_2 = 15$$

a) 1.561 V b) 1.651 V c) 1.732 V d) 2.348 V

**Q4** For the PMOS transistor given in the figure below, consider:
$I_D$ = 0.2mA, $V_{SD}$ =3 Volts, $V_{TP}$ = –2V, $k_p'$ = 25 μA/V², L = 3.5 μm, λ = 0.




What will be the value of W(width) of the transistor? _____ (in μm)

**Q5** If the pinch-off voltage for a Junction Field Effect Transistor (JFET) is 4 volts, what is the cut-off voltage for that?
a) 3 V b) 3.5 V c) 4 V d) 4.5 V

**Answer Key:**
1) b) 2) a) 3) d) 4) 56 μm 5) c)